64 mm (2.52 in) bore × 40 mm (1.57 in) rod Cylinder Force Data

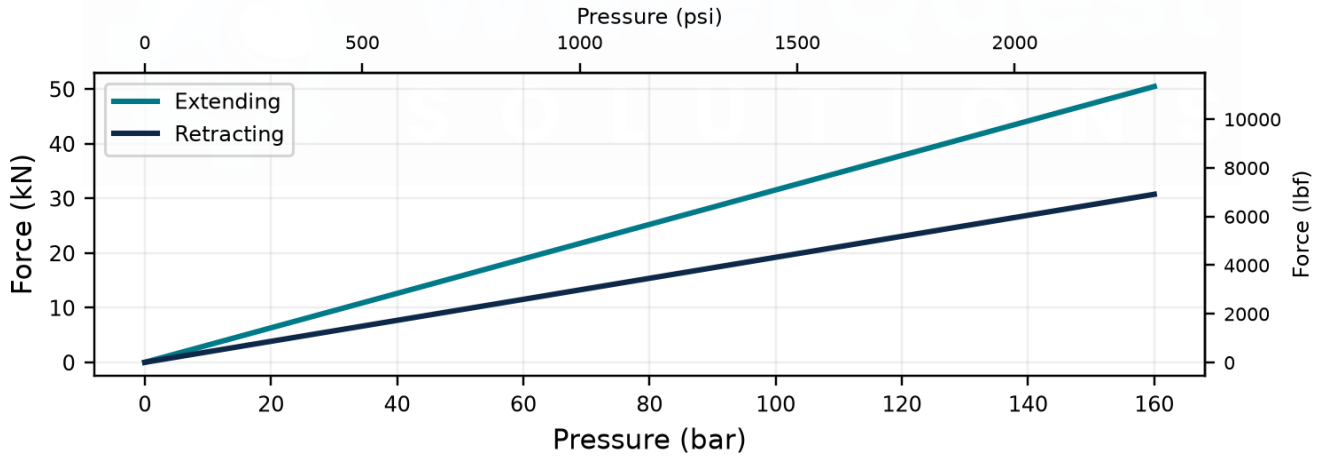
This geometry has a retraction area equal to 60.9% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 32.17 cm² (4.986 in²)	Ring Area 19.6 cm² (3.038 in²)
Max Extending Force 50.44 kN (11,339 lbf)	Max Retracting Force 30.73 kN (6,908 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	3.15 kN (708 lbf)	1.92 kN (432 lbf)
20 bar (290 psi)	6.31 kN (1,419 lbf)	3.84 kN (863 lbf)
30 bar (435 psi)	9.46 kN (2,127 lbf)	5.76 kN (1,295 lbf)
40 bar (580 psi)	12.61 kN (2,835 lbf)	7.68 kN (1,727 lbf)
50 bar (725 psi)	15.76 kN (3,543 lbf)	9.60 kN (2,158 lbf)
60 bar (870 psi)	18.92 kN (4,253 lbf)	11.52 kN (2,590 lbf)
70 bar (1,015 psi)	22.07 kN (4,962 lbf)	13.45 kN (3,024 lbf)
80 bar (1,160 psi)	25.22 kN (5,670 lbf)	15.37 kN (3,455 lbf)
90 bar (1,305 psi)	28.37 kN (6,378 lbf)	17.29 kN (3,887 lbf)
100 bar (1,450 psi)	31.53 kN (7,088 lbf)	19.21 kN (4,319 lbf)
110 bar (1,595 psi)	34.68 kN (7,796 lbf)	21.13 kN (4,750 lbf)
120 bar (1,740 psi)	37.83 kN (8,505 lbf)	23.05 kN (5,182 lbf)
130 bar (1,885 psi)	40.98 kN (9,213 lbf)	24.97 kN (5,613 lbf)
140 bar (2,031 psi)	44.14 kN (9,923 lbf)	26.89 kN (6,045 lbf)
150 bar (2,176 psi)	47.29 kN (10,631 lbf)	28.81 kN (6,477 lbf)
160 bar (2,321 psi)	50.44 kN (11,339 lbf)	30.73 kN (6,908 lbf)